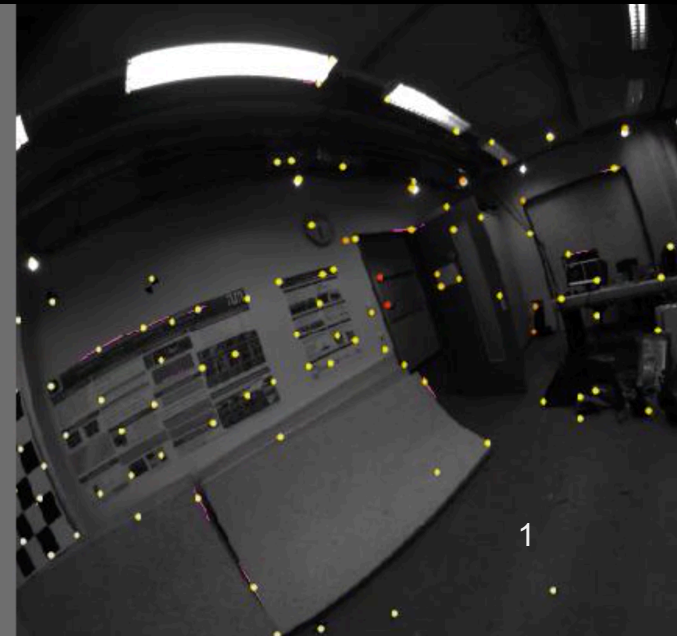
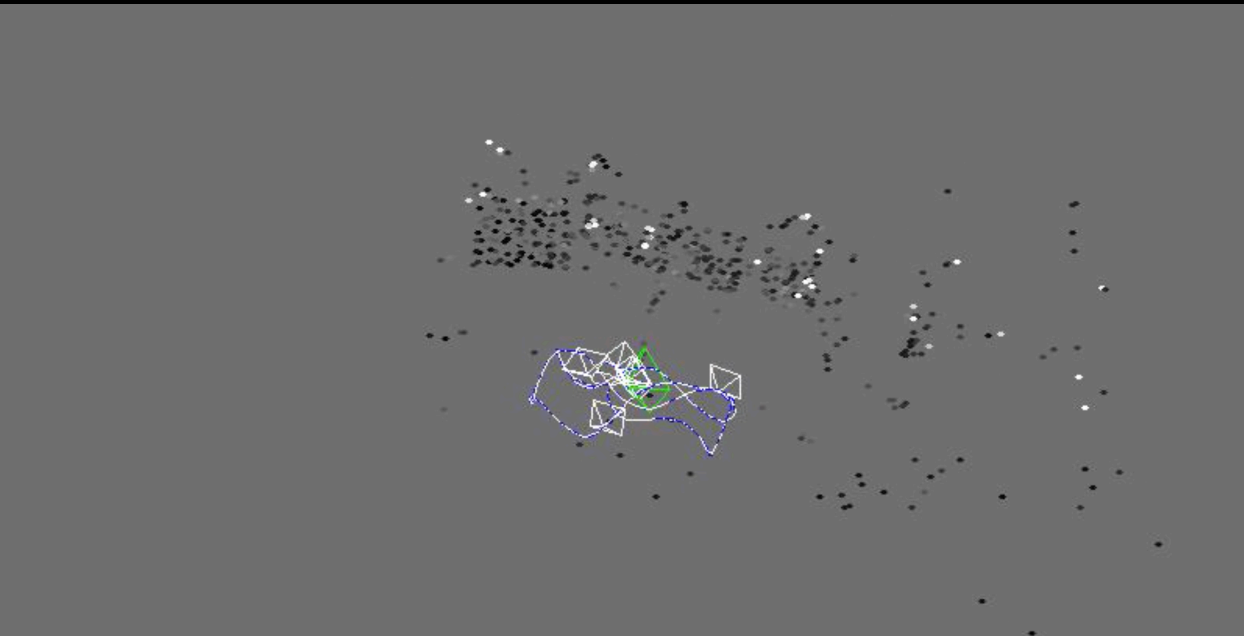


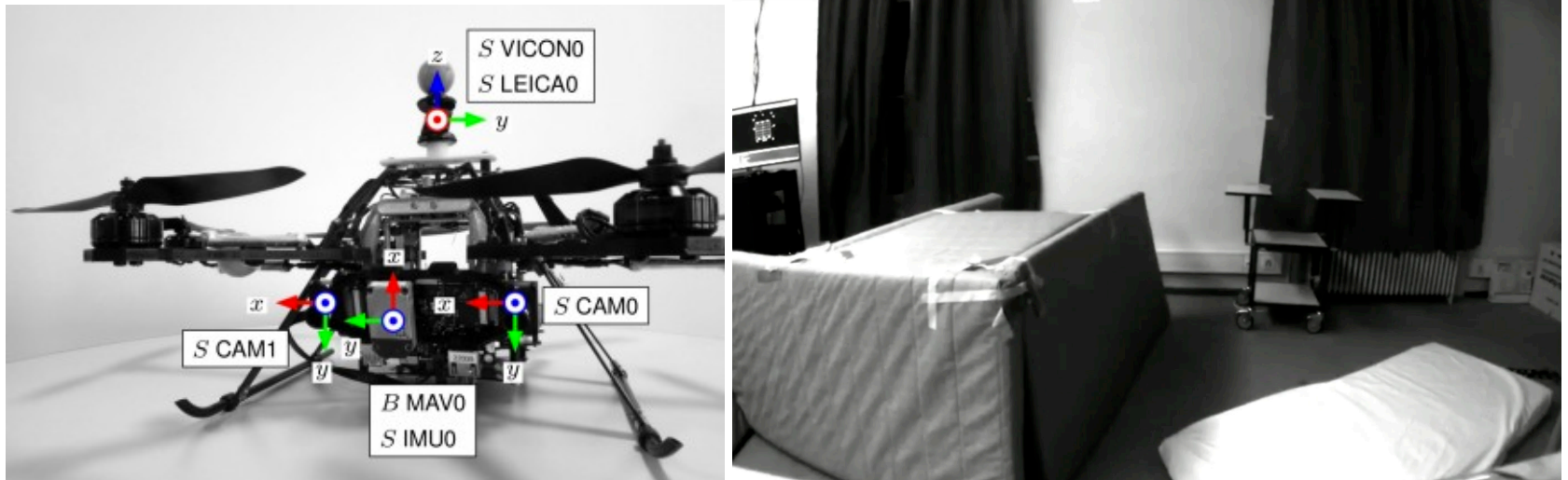
A QUASI-DIRECT APPROACH TO MONOCULAR VISUAL ODOMETRY

Purdue University School of Aeronautics and Astronautics

Kevin Sheridan



WHAT IS MONOCULAR VISUAL ODOMETRY?



Source: M. Burri, J. Nikolic, P. Gohl, T. Schneider, J. Rehder, S. Omari, M. Achtelik and R. Siegwart, *The EuRoC micro aerial vehicle datasets*, **International Journal of Robotic Research**

Goal:

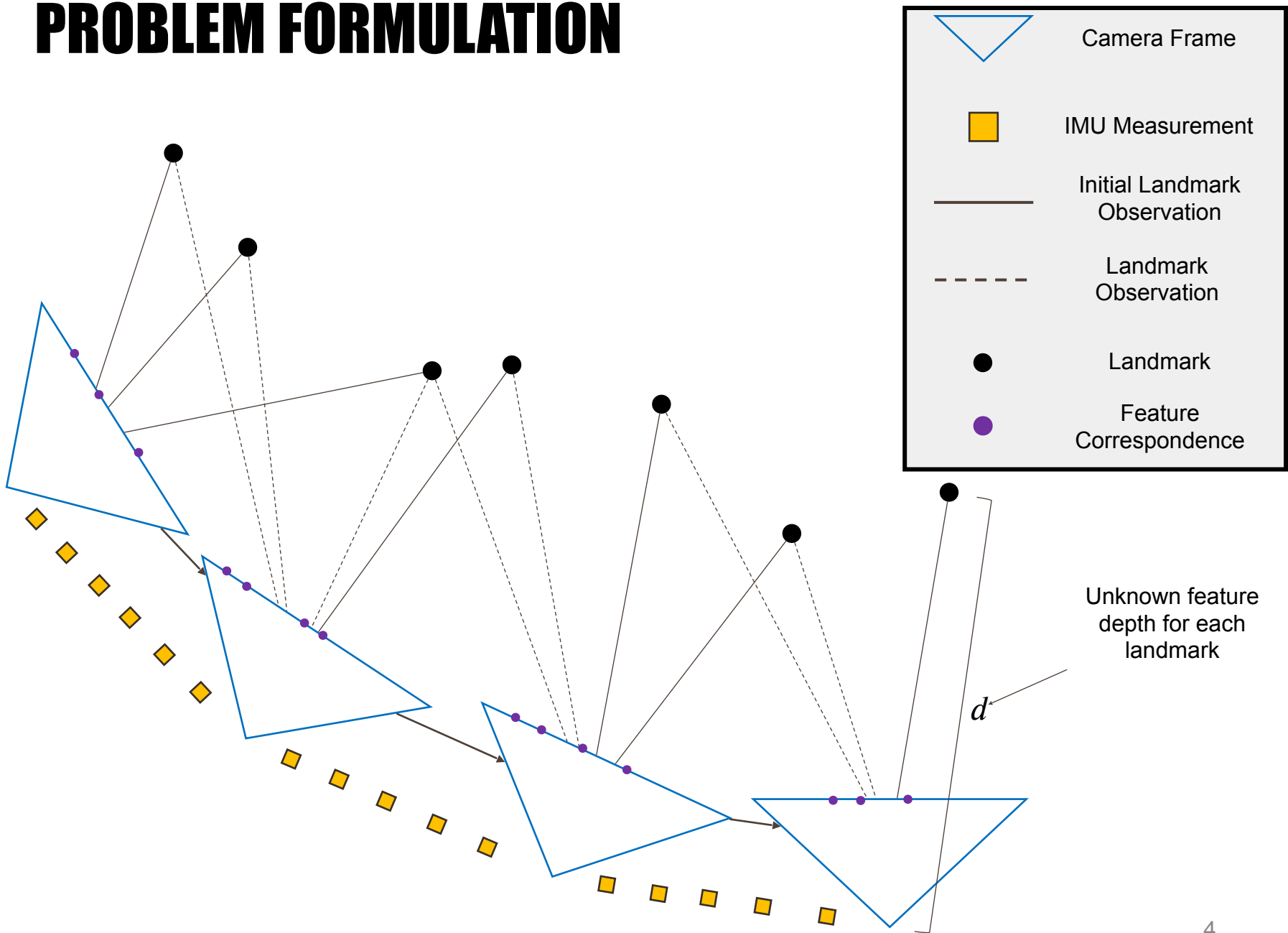
Using only one camera, estimate the motion of the camera.

Scene structure estimate is a byproduct.

DEMO



PROBLEM FORMULATION



INDIRECT APPROACH

$$\mathbf{x} \in SE(3)^n \times \mathbb{R}^m$$

Our state, $\mathbf{x}$, consists of n camera poses and m landmark inverse depths, d_{inv} .

Assuming we have matched features correctly

$$\mathbf{z}_{i,j,k} \in \mathcal{Z} \subset \mathbb{R}^2$$

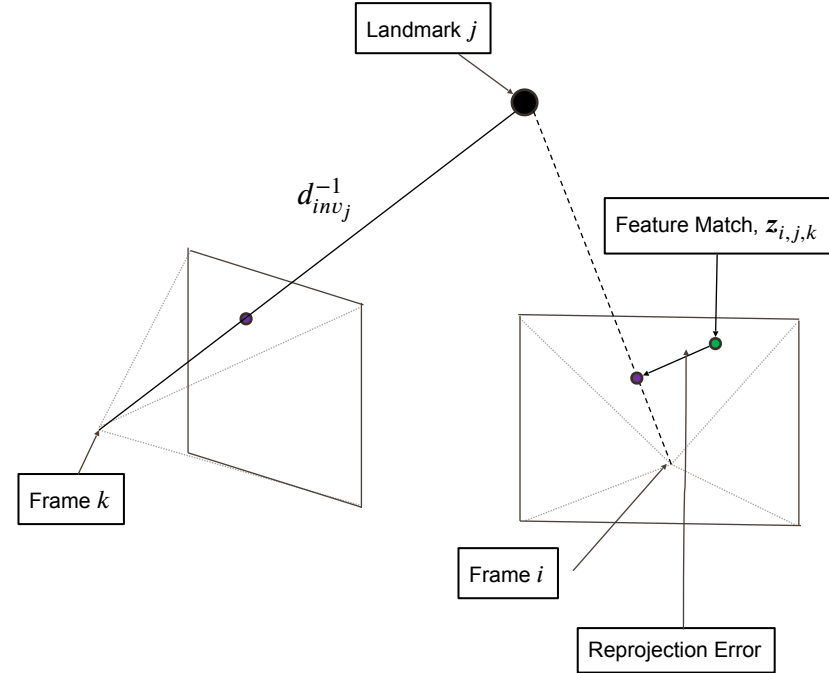
$\mathbf{z}_{i,j,k}$ is the pixel feature correspondence in frame i of landmark j which originated in frame k .

$$\begin{bmatrix} \mathbf{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix} = \mathbf{T}_i^{-1} \mathbf{T}_k \quad \mathbf{T} \in SE(3)$$

$$E_{reprojection} = \sum_{\mathbf{z}_{i,j,k} \in \mathcal{Z}} \left\| \Pi \left(\mathbf{R} \mathbf{u}_{0j} d_{inv_j}^{-1} + \mathbf{t} \right) - \mathbf{z}_{i,j,k} \right\|_{\Sigma_{correspondence}}^2$$

$$\Pi : \mathbb{R}^3 \mapsto \mathbb{R}^2$$

Π is a function which projects a 3D point into a pixel position. This handles lens distortion. $\mathbf{u}_0$ is the original homogenous undistorted feature bearing.



MAXIMUM A POSTERIORI ESTIMATION

$$\begin{aligned}\mathbf{x}^* &= \underset{\mathbf{x}}{\operatorname{argmax}} \quad p(\mathbf{x} \mid \mathcal{Z}) \\ p(\mathbf{x} \mid \mathcal{Z}) &\propto p(\mathbf{x}_0) \prod_{\mathbf{z}_{i,j,k} \in \mathcal{Z}} p(\mathbf{z}_{i,j,k} \mid \mathbf{x}) \\ \mathbf{x}^* &= \underset{\mathbf{x}}{\operatorname{argmin}} \quad -\log_e \left[p(\mathbf{x}_0) \prod_{\mathbf{z}_{i,j,k} \in \mathcal{Z}} p(\mathbf{z}_{i,j,k} \mid \mathbf{x}) \right] \\ p(\mathbf{z}_{i,j,k} \mid \mathbf{x}) &\propto e^{-\frac{1}{2} \left\| \Pi \left(\mathbf{R} \mathbf{u}_{0_j} d_{inv_j}^{-1} + \mathbf{t} \right) - \mathbf{z}_{i,j,k} \right\|_{\Sigma_{correspondence}}^2} \\ p(\mathbf{x}_0) &\propto e^{-\frac{1}{2} \left\| \mathbf{x} - \mathbf{x}_0 \right\|_{\Sigma_{prior}}^2}\end{aligned}$$

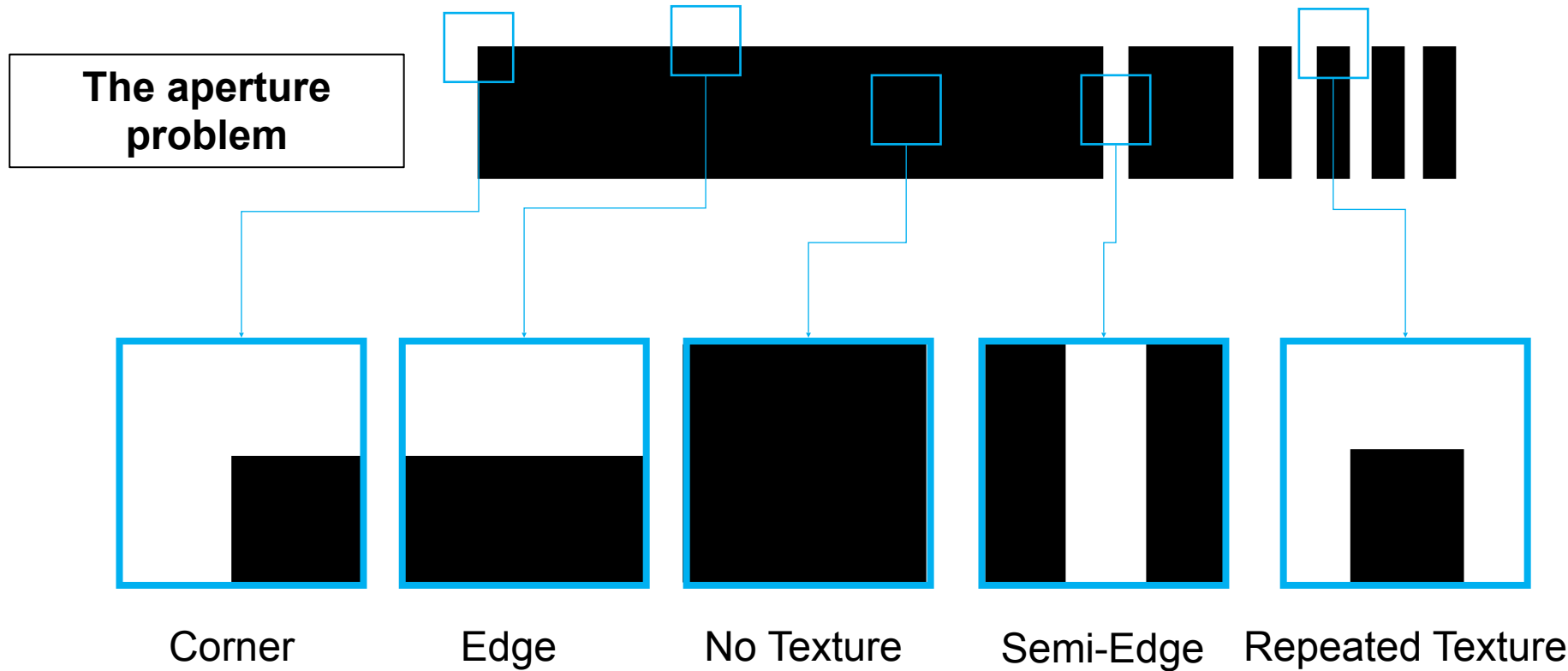
$$\boxed{\mathbf{x}^* = \underset{\mathbf{x}}{\operatorname{argmin}} \quad E_{prior} + E_{reprojection}}$$

This is known as an indirect method

THE PROBLEM WITH INDIRECT METHODS

- Indirect methods require that features be matched correctly between frames.
 - In practice, this is a challenging and error prone task.
- The requirement that feature correspondences be established limits the types of features which can be used.

FEATURE MATCHING AND WHY IT IS DIFFICULT



Indirect



Indirect Methods

Image In

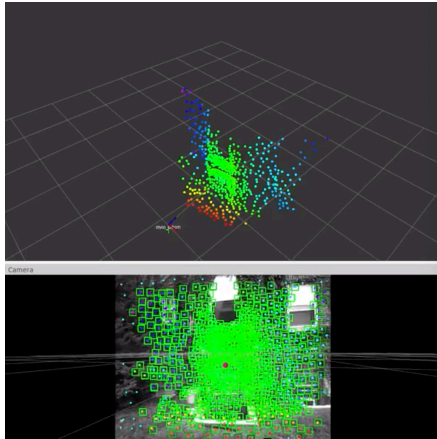


Feature Matching

Find map
landmarks in new
image.

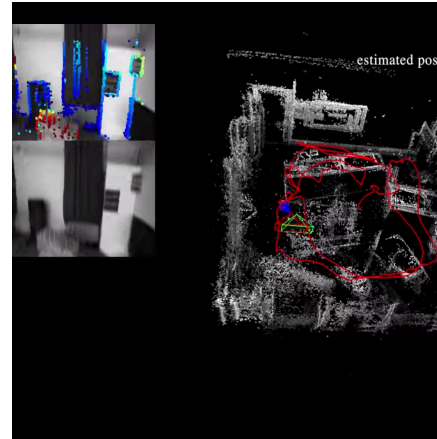


Determine
landmark position
and camera pose
by minimizing the
reprojection error.



Direct Methods

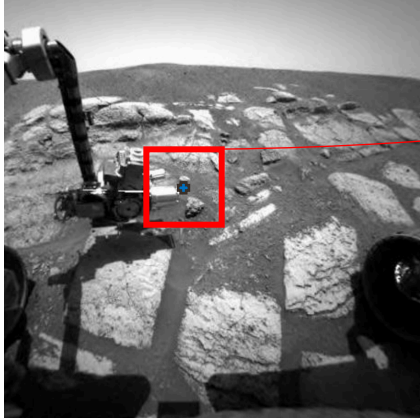
Image In



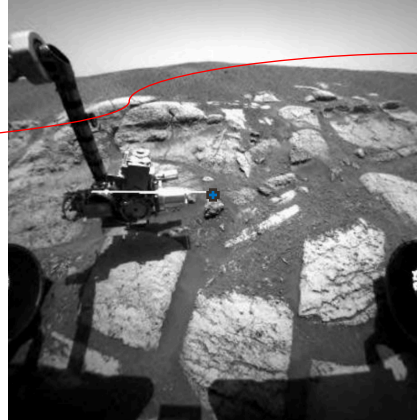
Determine
landmark position
and camera pose
by minimizing the
photometric error.

Image Source: VI-DSO

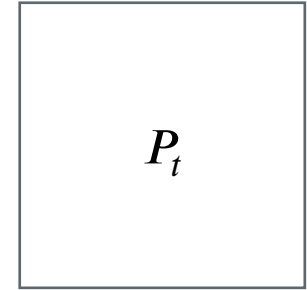
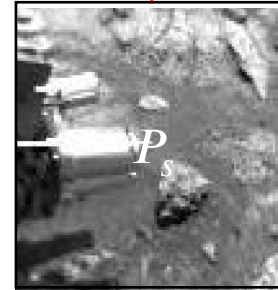
PHOTOMETRIC RESIDUALS



$$\text{intensity} = I_s(x, y)$$



$$\text{intensity} = I_t(x, y)$$



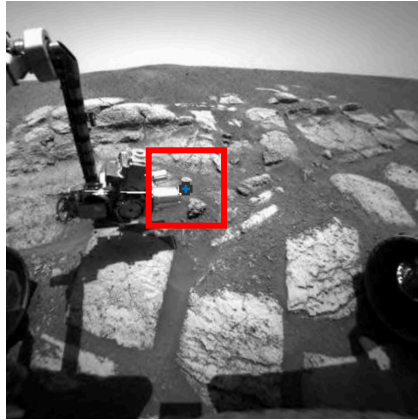
$$SSD(P_s, P_t) = \sum_{p \in R} \|P_s(p) - P_t(p)\|^2$$

$$\begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} = \mathbf{T}_i^{-1} \mathbf{T}_k$$

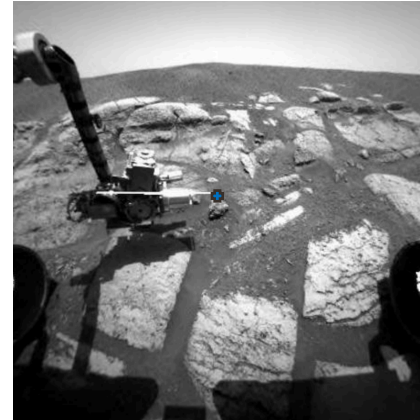
$$E_{\text{photometric}} = \sum_{i=1}^n \sum_{k=1}^n \sum_{p\mathbf{x}_0_j, d_{inv_j} \in \mathcal{N}_k} \left\| I_t \left[\Pi \left(R \Pi^{-1} (p\mathbf{x}_0_j, d_{inv_j}) + t \right) \right] - I_s(p\mathbf{x}_0_j) \right\|^2$$

$\mathcal{N}_k$ is the set of landmarks in frame k . $p\mathbf{x}_0$ is the first observed pixel position of a feature

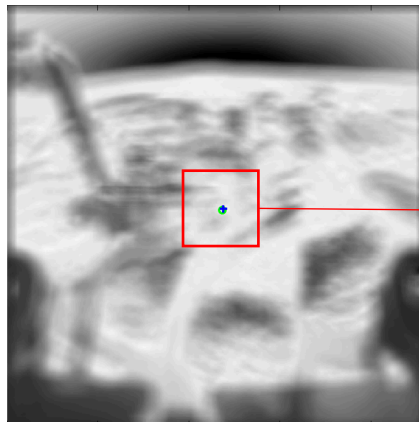
THE PROBLEM WITH DIRECT METHODS



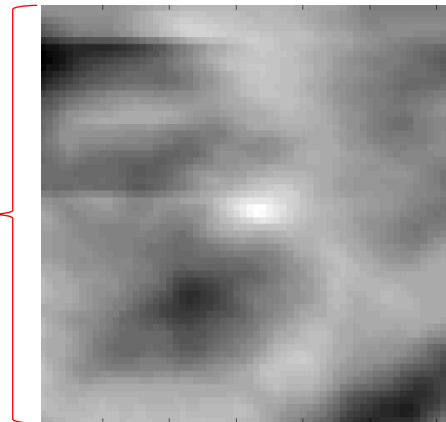
$$\text{intensity} = I_s(x, y)$$



$$\text{intensity} = I_t(x, y)$$



SSD over I_t



Highly sensitive to illumination changes.

Highly non convex.

QUASI-DIRECT RESIDUAL

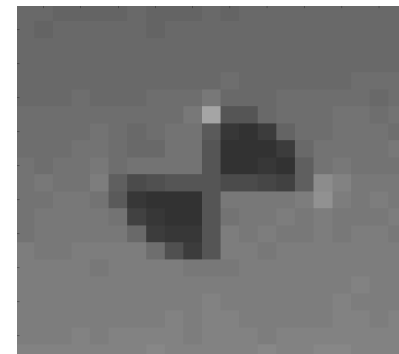
What do we want?

1. Illumination Invariant
2. Convex
3. Can handle the same feature types as a direct method.



Illumination Invariant Patch Comparison Metrics

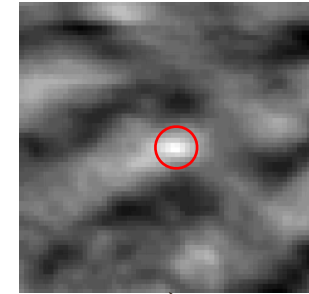
- Zero Mean Normalized Cross Correlation (ZNCC)
- ORB/BRIEF
- Fast Retina Keypoints (FREAK)
- BRISK
- ...



QUASI-DIRECT RESIDUAL

What do we want?

1. Illumination Invariant ✓
2. Convex
3. Can handle the same feature types as a direct method.



Typical ZNCC response for corner feature.

Pixel position $\leftarrow \begin{bmatrix} px_i \\ s_i \end{bmatrix} \in \mathcal{X}_j^k \subset \mathbb{Z}^2 \times \mathbb{R}$
 Score (ZNCC) $\leftarrow$

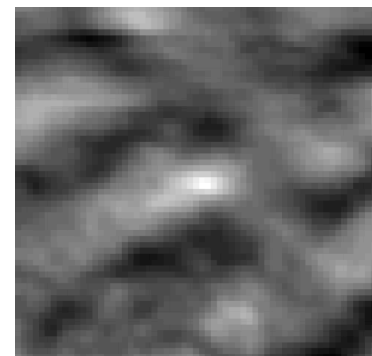
$$\{\theta \in \mathbb{R} : -1 \leq \theta \leq 1\}$$

Correspondence Distribution

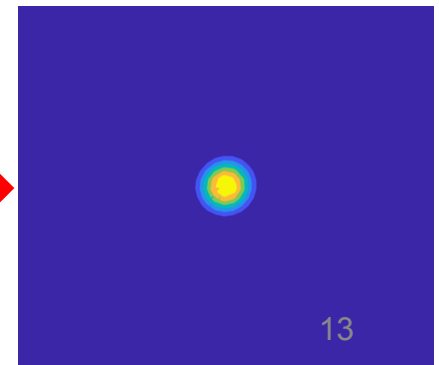
$$p(px | \mathcal{X}_j^k) = \sum_{i \in \mathcal{X}_j^k, \theta < s_i \leq 1} \hat{s}_i \mathcal{N}(px_i, \Sigma)$$

$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

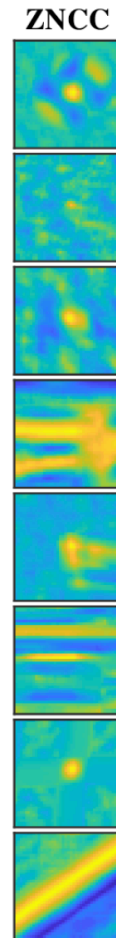
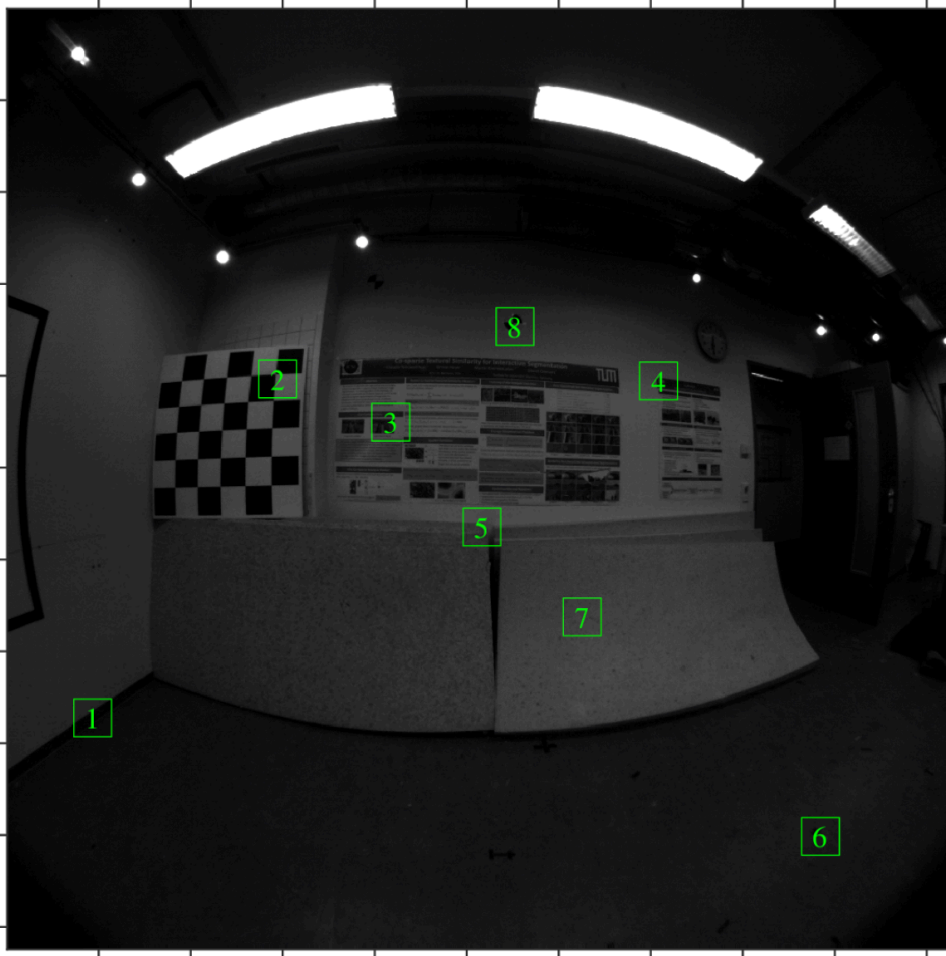
$$\hat{s}_i(s_i | \mathcal{X}_j^k) = \frac{s_i - \theta}{\sum_{k \in \mathcal{X}_j^k, \theta < s_k \leq 1} s_k - \theta}$$



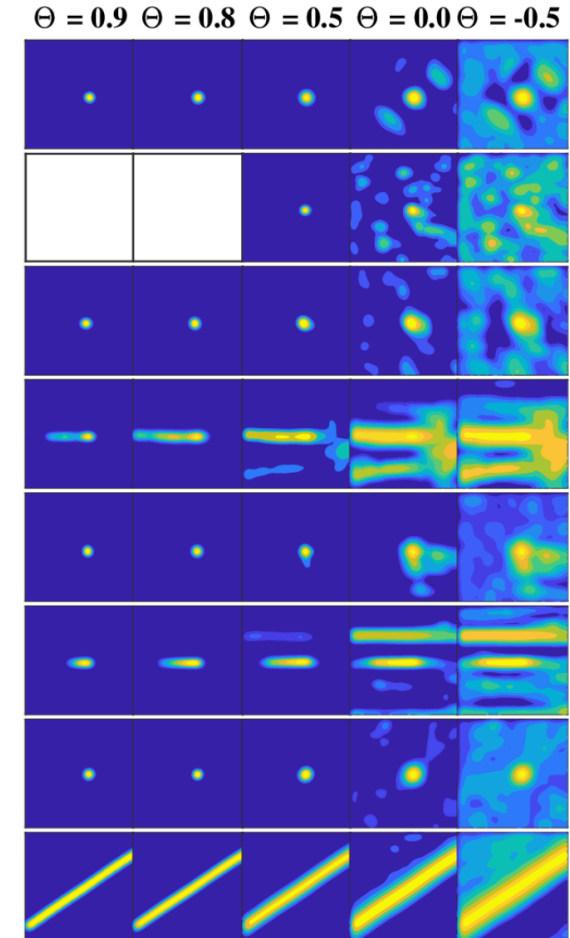
$\theta = 0.8$



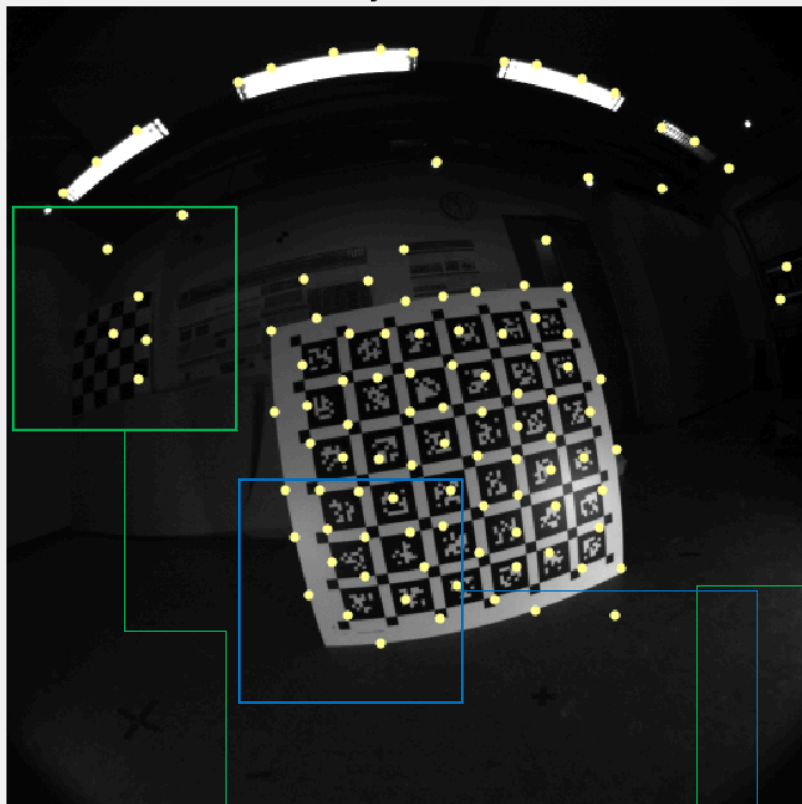
QUASI-DIRECT RESIDUAL



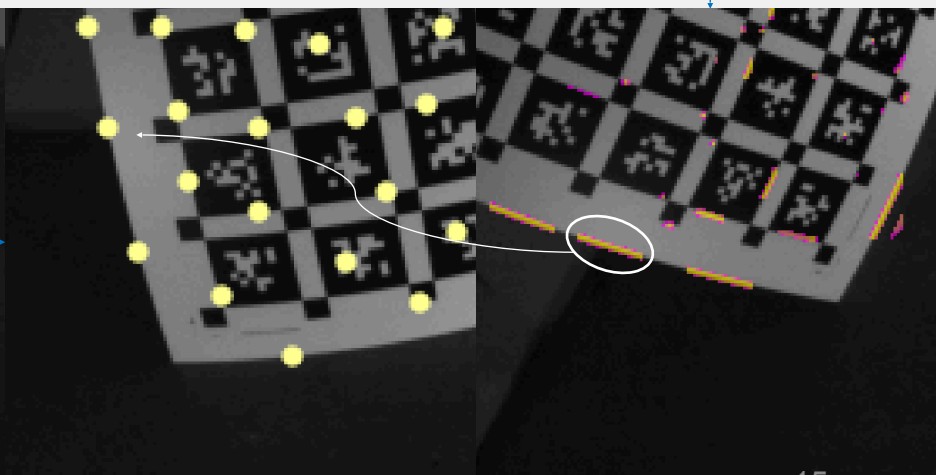
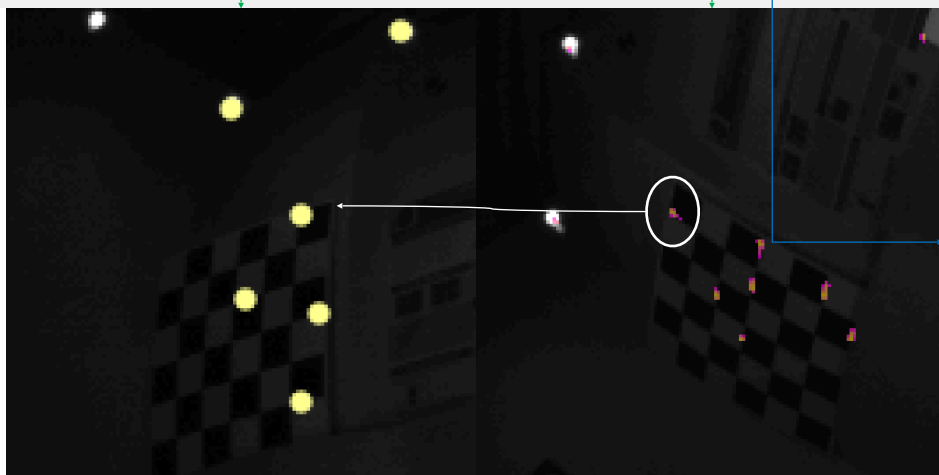
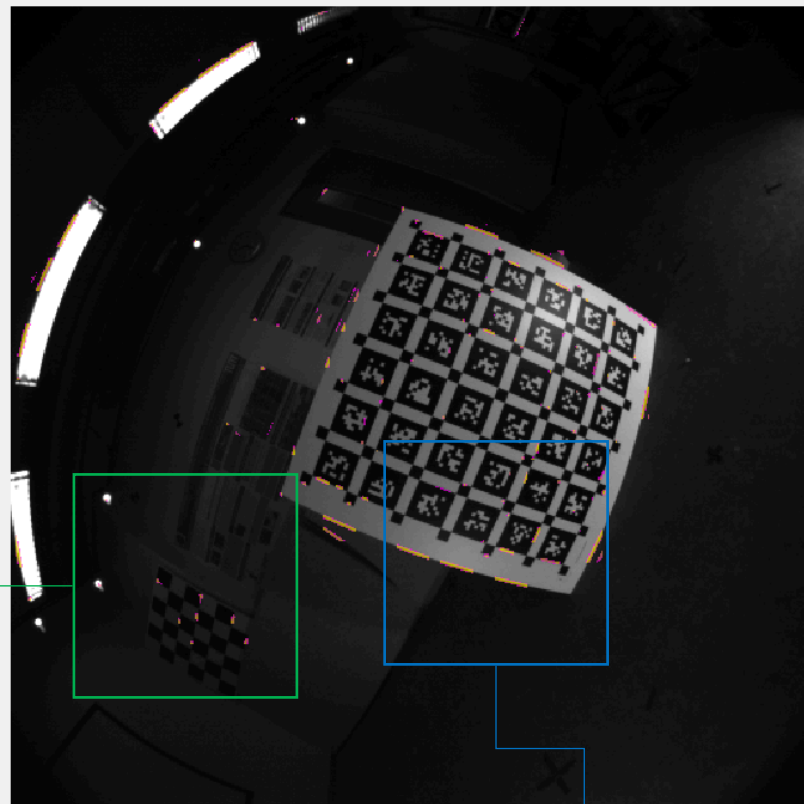
Gaussian Mixture Model Approximation



Keyframe 1



Frame 10



QUASI-DIRECT RESIDUAL

How can we efficiently optimize this correspondence model?

During maximum a posteriori estimation, we must minimize the negative log likelihood.

$$\mathbf{x}^* = \underset{\mathbf{x}}{\operatorname{argmin}} \quad -\log_e \left[p(\mathbf{x}_0) \prod_{\mathbf{z}_{i,j,k} \in \mathcal{Z}} p(\mathbf{z}_{i,j,k} | \mathbf{x}) \right]$$

Unlike direct and indirect methods, the negative log likelihood of a gaussian mixture model does not simplify to a sum of squared error.

$$-\log_e \left[p(px | \mathcal{Z}_j^k) \right] = -\log_e \left[\sum_{i \in \mathcal{Z}_j^k, \theta < s_i \leq 1} \hat{s}_i \mathcal{N}(px_i, \Sigma) \right]$$

$$\begin{bmatrix} \mathbf{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix} = \mathbf{T}_i^{-1} \mathbf{T}_k \quad p(px | \mathcal{Z}_j^k) \propto \sum_{i \in \mathcal{Z}_j^k, \theta < s_i \leq 1} \hat{s}_i e^{-\frac{1}{2} \left\| \Pi(\mathbf{R} \mathbf{u}_{0_j} d_{inv_j}^{-1} + \mathbf{t}) - px_i \right\|^2}$$

$$\left\| \nabla_{px} [-\log_e p(px | \mathcal{Z}_j^k)] \right\| = \sum_{i \in \mathcal{Z}_j^k, \theta < s_i \leq 1} \left[\frac{\hat{s}_i \mathcal{N}(px_i, \Sigma)}{\sum_{l \in \mathcal{Z}_j^k, \theta < s_l \leq 1} \hat{s}_l \mathcal{N}(px_l, \Sigma)} \right] \left\| \Pi(\mathbf{R} \mathbf{u}_{0_j} d_{inv_j}^{-1} + \mathbf{t}) - px_i \right\| = 0$$

The gradient magnitude of the correspondence distribution is a weighted reprojection error.

QUASI-DIRECT RESIDUAL

$$\nabla_{px}[-\log_e p(px | \mathcal{Z}_j^k)] = \sum_{i \in \mathcal{Z}_j^k, \theta < s_i \leq 1} \frac{\hat{s}_i \mathcal{N}(px_i, \Sigma)}{\sum_{l \in \mathcal{Z}_j^k, \theta < s_l \leq 1} \hat{s}_l \mathcal{N}(px_l, \Sigma)} [\Pi(Ru_{0j} d_{inv_j}^{-1} + t) - px_i] = 0$$

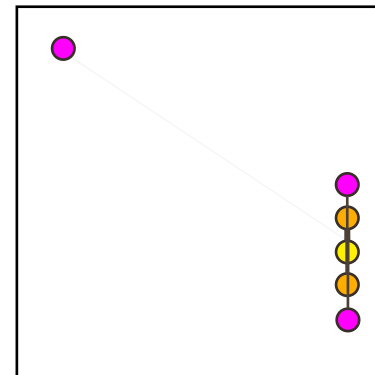
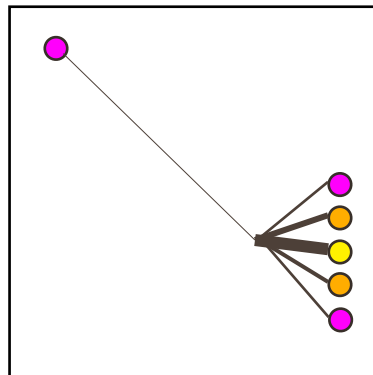
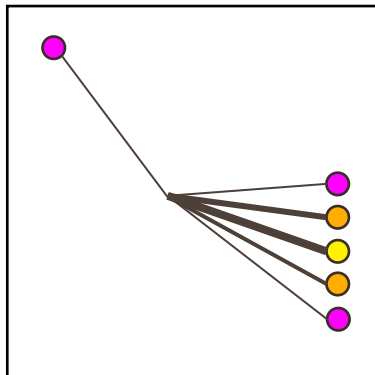
Gaussian Weights

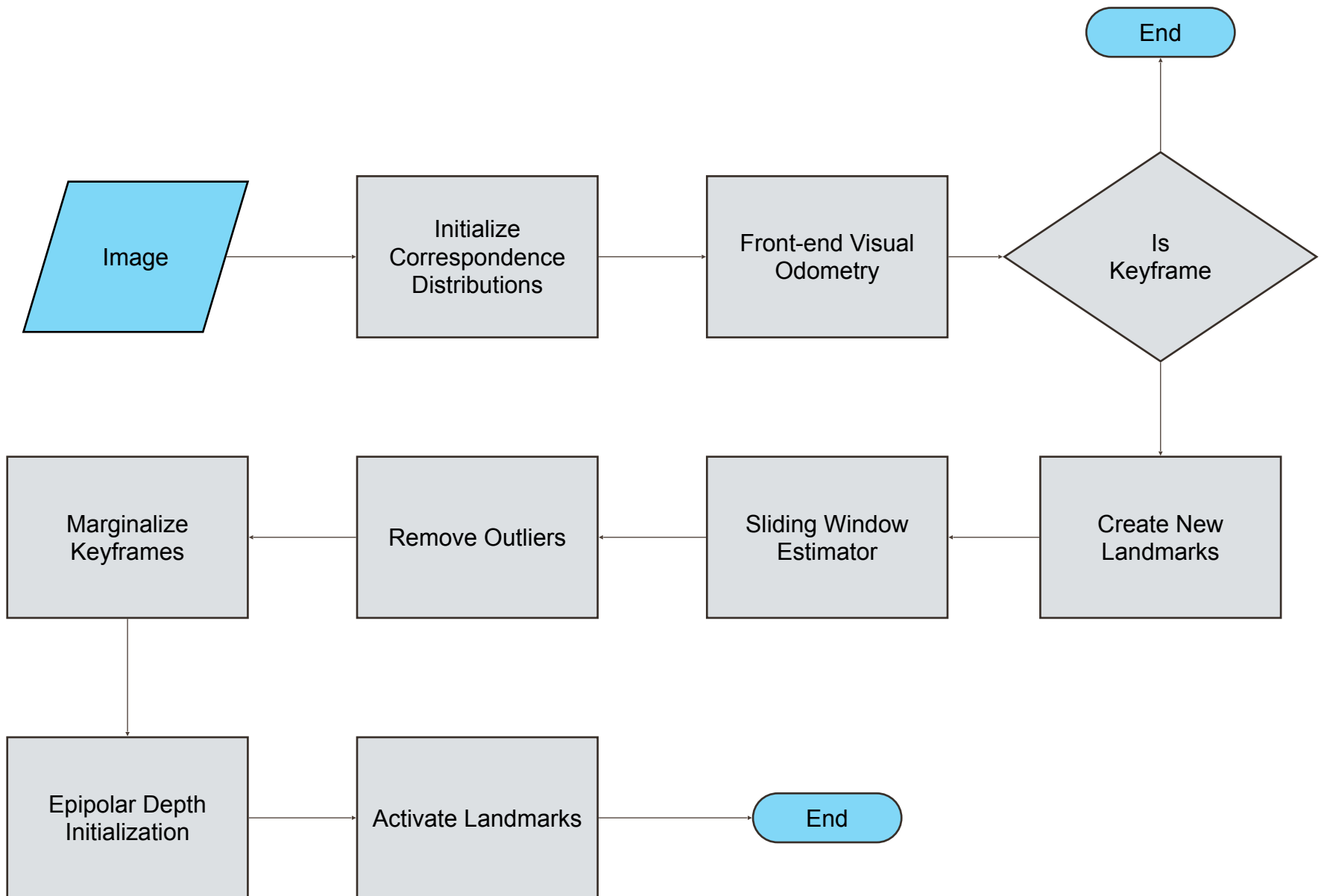
If we assume these weights are constant, the optimization problem can be solved using standard quasi newton methods like Levenberg-Marquardt.

$$\frac{\hat{s}_i \mathcal{N}(px_i, \Sigma)}{\sum_{l \in \mathcal{Z}_j^k, \theta < s_l \leq 1} \hat{s}_l \mathcal{N}(px_l, \Sigma)}$$

Iteratively Reweighted Nonlinear Least Squares

θ  1





RESULTS

CONCLUSION AND FUTURE WORK

Conclusion

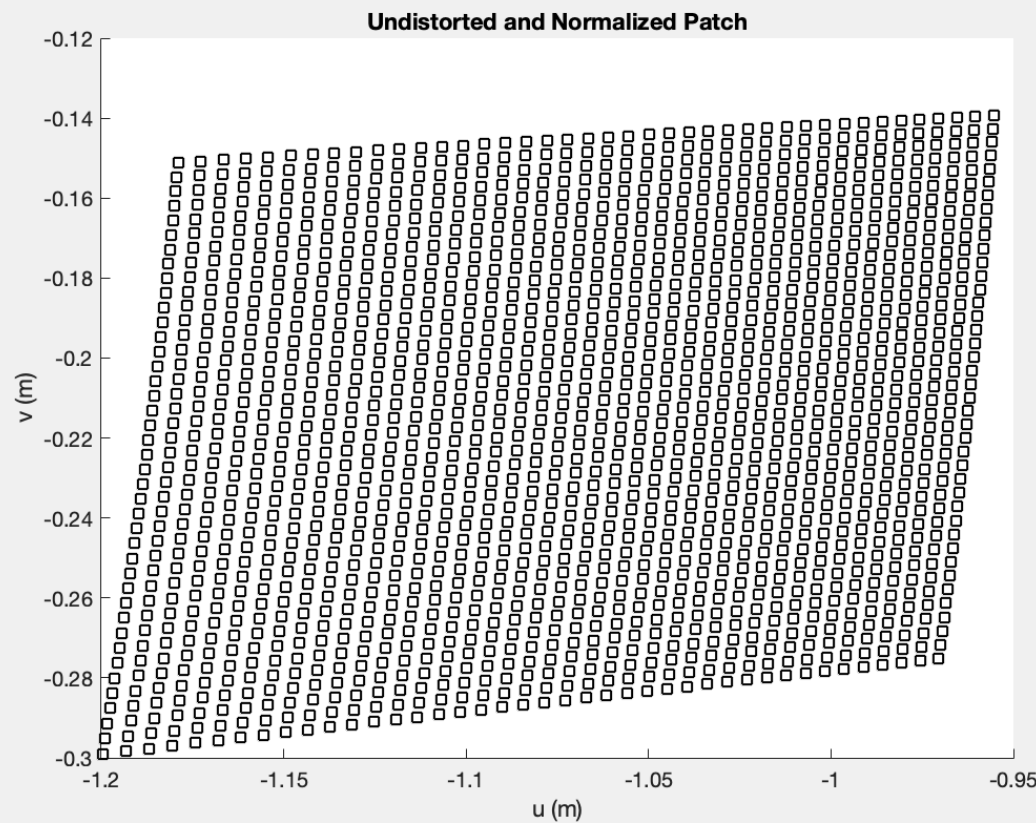
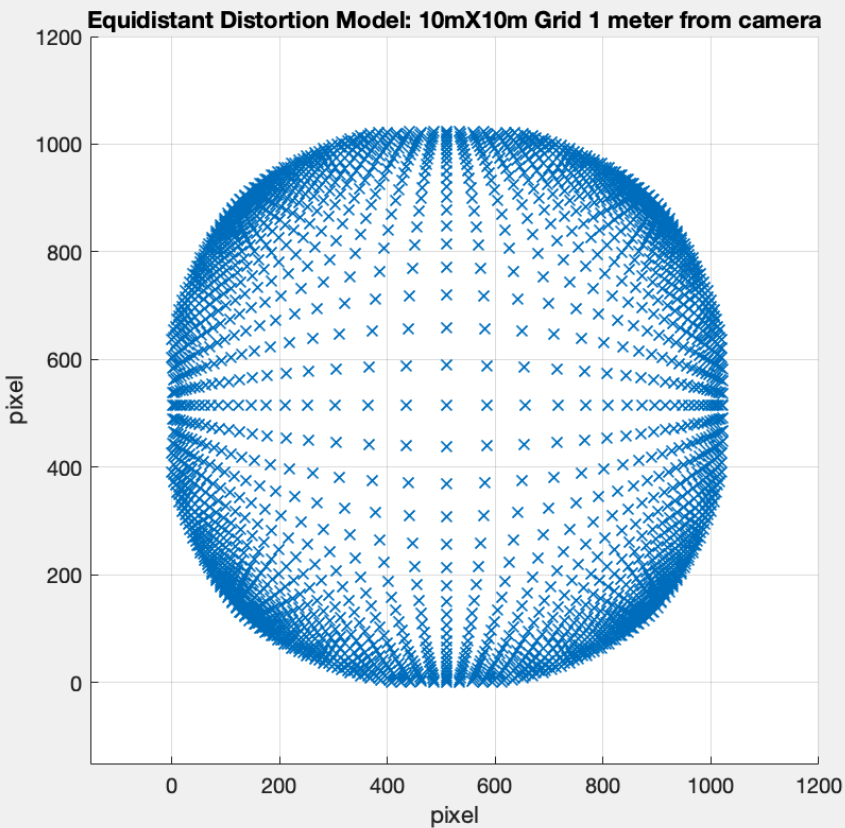
- Low Drift Achieved
- Edges Reconstructed
- Convex Residual allows for aggressive motion

Future Work

- Quantitative Drift Results
- Real time C++ Implementation
- Evaluate more patch comparisons
- Adaptive Thresholding

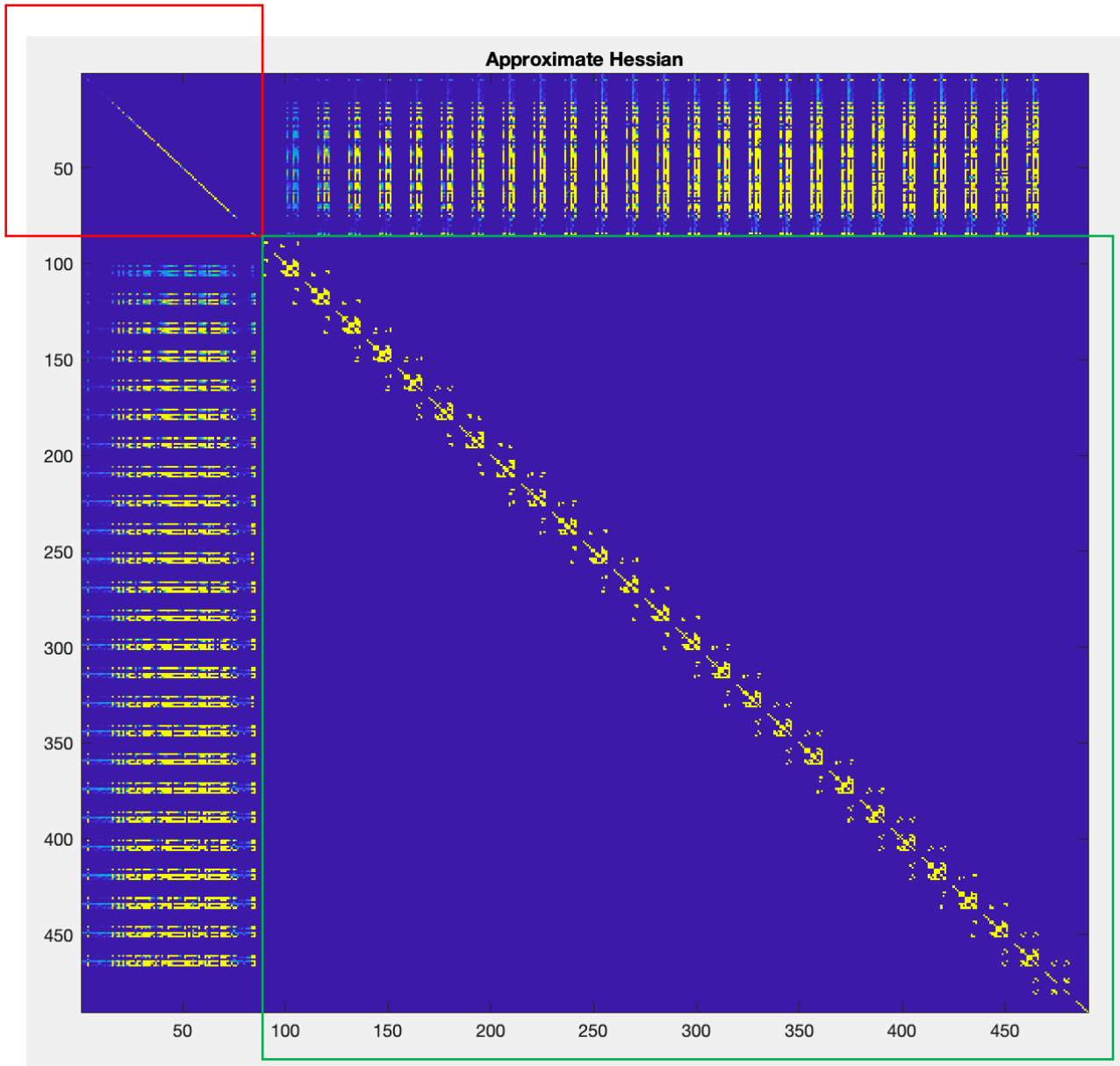
QUESTIONS

EQUIDISTANT LENS DISTORTION MODEL



HESSIAN APPROXIMATION

Landmark
Depths



Camera Poses

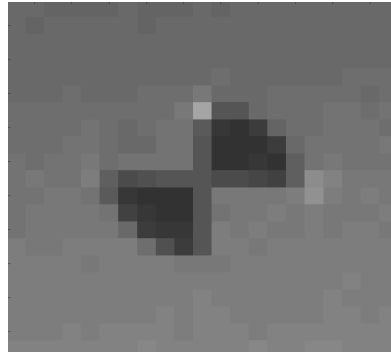
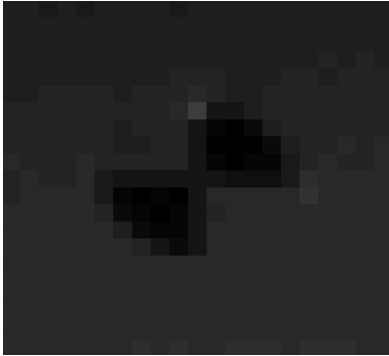
ZNCC

Unknown Illumination Change

$$P_t = aP_s + b$$

P_s - Source Patch

P_t - Target Patch



$$\bar{P}_t = a\bar{P}_s + b$$

$$P_t - \bar{P}_t = aP_s - a\bar{P}_s = a(P_s - \bar{P}_s)$$

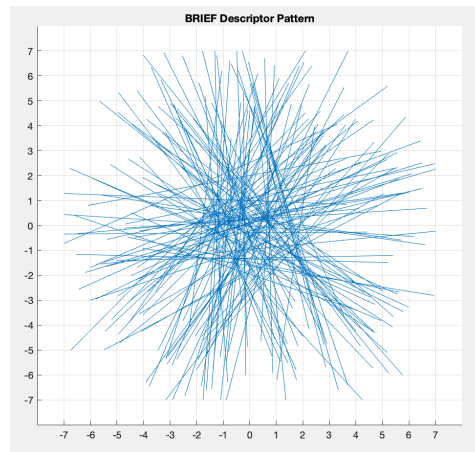
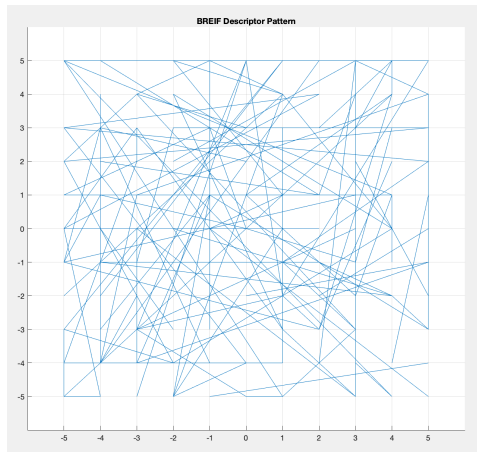
$$sd(P_t - \bar{P}_t) = a * sd(P_s - \bar{P}_s)$$

$$\hat{P}_s = \frac{P_s - \bar{P}_s}{\sqrt{\sum (P_s - \bar{P}_s)^2}} \quad \hat{P}_t = \frac{P_t - \bar{P}_t}{\sqrt{\sum (P_t - \bar{P}_t)^2}} \quad \leftarrow \text{Patch Normalization}$$

$$ZNCC(P_s, P_t) = \sum_{p \in R} P_s(p) P_t(p)$$

$$-1 \geq ZNCC(\dots) \leq 1$$

BRIEF - BINARY FEATURE DESCRIPTORS



Definition:

$$\tau(\mathbf{p}; \mathbf{x}, \mathbf{y}) := \begin{cases} 1 & \text{if } \mathbf{p}(\mathbf{x}) < \mathbf{p}(\mathbf{y}) \\ 0 & \text{otherwise} \end{cases}$$

$$f_{n_d}(\mathbf{p}) := \sum_{1 \leq i \leq n_d} 2^{i-1} \tau(\mathbf{p}; \mathbf{x}_i, \mathbf{y}_i)$$

Illumination Invariant: $c < d \Rightarrow ac + b < ad + b$

